PES University

Department of Computer Science and Engineering

Machine Learning Lab

Week 10: Support Vector Machine (SVM) Classifier

Lab Report

Name	_____
SRN	PES2UG24CS616
Section	____

1. Analysis Questions

Moons Dataset

Q1. Based on the metrics and the visualizations, what inferences about the performance of the Linear Kernel can you draw?

The Linear kernel reached 87% accuracy (F1 of 0.87 for class 0 and 0.86 for class 1), the lowest of the three kernels on this dataset. The plot shows why: it can only draw a single straight line, while the two moons interlock in a curve. The line cuts across the tips of both moons, so the points at the inner end of each moon fall on the wrong side. The model is therefore underfitting; it captures the general top-left versus bottom-right trend but not the actual shape, and no choice of straight line could fix this because the data is not linearly separable.

Q2. Compare the decision boundaries of the RBF and Polynomial kernels. Which one seems to capture the shape of the data more naturally?

The RBF kernel captures the shape far more naturally. Its boundary is a smooth S-shaped curve that bends around each moon and follows the gap between them, giving 97% accuracy with recall of 1.00 on class 0. The Polynomial kernel (default degree 3) produces a boundary that is almost a straight line with only a slight bend near the centre, so it behaves much like the linear model and reaches only 89% accuracy, with recall for class 1 dropping to 0.83. RBF is a local, distance-based kernel, so it can adapt to the curvature in each region, whereas a cubic polynomial imposes one global shape that does not match two interlocking arcs.

Banknote Dataset

Q1. In this case, which kernel appears to be the most effective?

The RBF kernel is the most effective, with 93% accuracy against 88% for Linear and 84% for Polynomial. It also has the best balance across classes: F1 of 0.94 for Forged and 0.93 for Genuine, with recall of 0.96 on the Genuine class. Its curved boundary wraps around the cluster of one class and separates the overlapping central region better than the other two kernels.

Q2. The Polynomial kernel shows lower performance here compared to the Moons dataset. What might be the reason for this?

Using only variance and skewness, the two classes overlap heavily in the centre and the true boundary is irregular rather than a clean polynomial curve. The default polynomial kernel in scikit-learn is degree 3 with coef0 = 0, which makes it an odd function that is symmetric about the origin of the scaled feature space. That rigid, global shape cannot bend locally to follow the overlap, so the boundary cuts through the Genuine cluster and its recall falls to 0.75. It performs worse than even the Linear kernel (84% versus 88%), which shows the extra curvature is being placed in the wrong regions. Tuning degree, coef0 and C would likely improve it, but with defaults the kernel's shape simply does not match this data.

Hard vs. Soft Margin

Q1. Which model, the Soft Margin (C=0.1) or the Hard Margin (C=100), produces a wider margin?

The Soft Margin model (C = 0.1) produces the wider margin. A small C puts little penalty on margin violations, so the optimiser prefers a small weight vector, and since the margin width is $2/||w||$, the margin becomes wide. With C = 100 violations are penalised heavily, so the model shrinks the margin to classify as many training points correctly as possible.

Q2. Why does the soft margin SVM allow these "mistakes"? What is the primary goal of this model?

With a small C the cost of a misclassified or in-margin point (its slack variable) is low compared to the benefit of a larger margin. The model's primary goal is not to get every training point right but to find the widest, most stable separation between the bulk of the two classes. It therefore treats the two added outliers as noise and ignores them instead of bending the boundary to accommodate them, which gives better generalisation.

Q3. Which of these two models is more likely to be overfitting to the training data? Explain your reasoning.

The Hard Margin model (C = 100) is more likely to be overfitting. Because every violation is penalised heavily, its boundary is pulled by individual points, including the outliers, and in the plot its boundary has clearly rotated to an almost horizontal line compared with the soft margin one. A boundary that depends on one or two noisy points has high variance: a small change in the training data would move it noticeably.

Q4. Which model do you trust more for a new, unseen data point? Which value of C would you generally prefer to start with for noisy data?

I would trust the Soft Margin model (C = 0.1) more. Its wider margin leaves more room on both sides of the boundary, so a new point that falls slightly away from the training clusters is still likely to be classified correctly, and the boundary reflects the overall structure of the data rather than its outliers. For noisy real-world data I would start with a low C (around 0.1 to 1) and then tune it with cross-validation, increasing it only if the model shows signs of underfitting.

2. Screenshots

Training Results: Moons Dataset

Screenshot 1: Classification Report for SVM with LINEAR Kernel (Moons)

```
SVM with LINEAR Kernel PES2UG24CS616
              precision    recall  f1-score   support

           0       0.85      0.89      0.87        75
           1       0.89      0.84      0.86        75

    accuracy                           0.87       150
   macro avg       0.87      0.87      0.87       150
weighted avg       0.87      0.87      0.87       150
```

Screenshot 2: Classification Report for SVM with RBF Kernel (Moons)

```
SVM with RBF Kernel PES2UG24CS616
              precision    recall  f1-score   support

           0       0.95      1.00      0.97        75
           1       1.00      0.95      0.97        75

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

Screenshot 3: Classification Report for SVM with POLY Kernel (Moons)

```
SVM with POLY Kernel PES2UG24CS616
              precision    recall  f1-score   support

           0       0.85      0.95      0.89        75
           1       0.94      0.83      0.88        75

    accuracy                           0.89       150
   macro avg       0.89      0.89      0.89       150
weighted avg       0.89      0.89      0.89       150
```

Training Results: Banknote Dataset

Screenshot 4: Classification Report for SVM with LINEAR Kernel (Banknote)

```
SVM with LINEAR Kernel PES2UG24CS616
              precision    recall  f1-score   support

      Forged       0.90      0.88      0.89       229
     Genuine       0.86      0.88      0.87       183

    accuracy                           0.88       412
   macro avg       0.88      0.88      0.88       412
weighted avg       0.88      0.88      0.88       412
```

Screenshot 5: Classification Report for SVM with RBF Kernel (Banknote)

```
SVM with RBF Kernel PES2UG24CS616
              precision    recall  f1-score   support

      Forged       0.96      0.91      0.94       229
     Genuine       0.90      0.96      0.93       183

    accuracy                           0.93       412
   macro avg       0.93      0.93      0.93       412
weighted avg       0.93      0.93      0.93       412
```

Screenshot 6: Classification Report for SVM with POLY Kernel (Banknote)

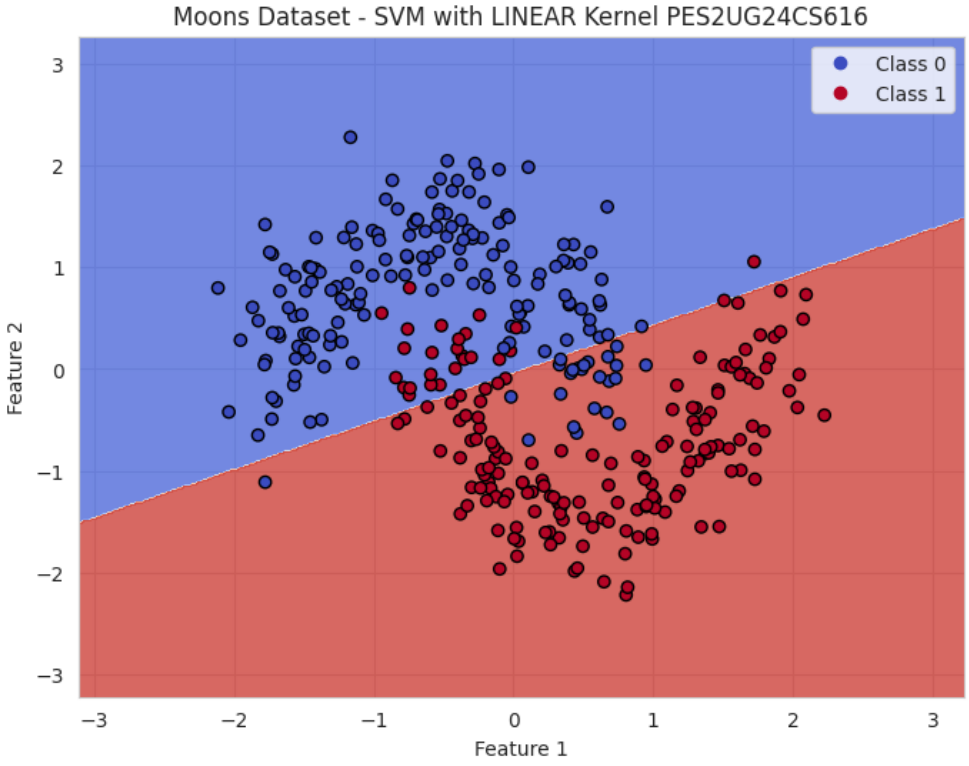
```
SVM with POLY Kernel PES2UG24CS616
              precision    recall  f1-score   support

      Forged       0.82      0.91      0.87       229
     Genuine       0.87      0.75      0.81       183

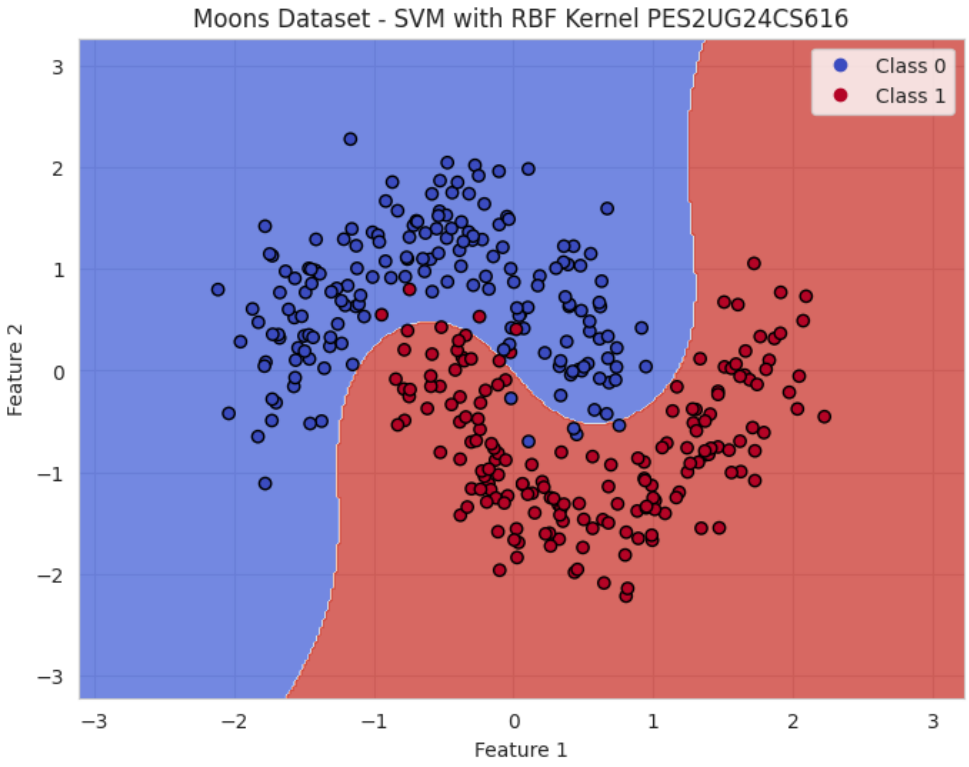
    accuracy                           0.84       412
   macro avg       0.85      0.83      0.84       412
weighted avg       0.85      0.84      0.84       412
```

Decision Boundary Visualizations

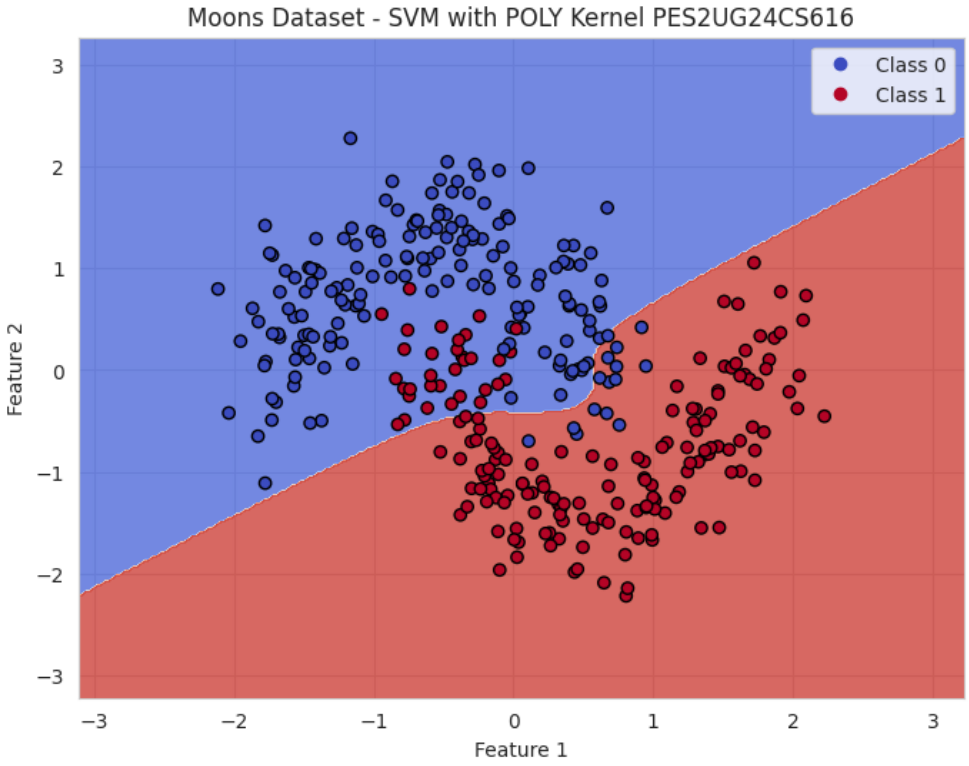
Screenshot 7: Moons Dataset - SVM with LINEAR Kernel



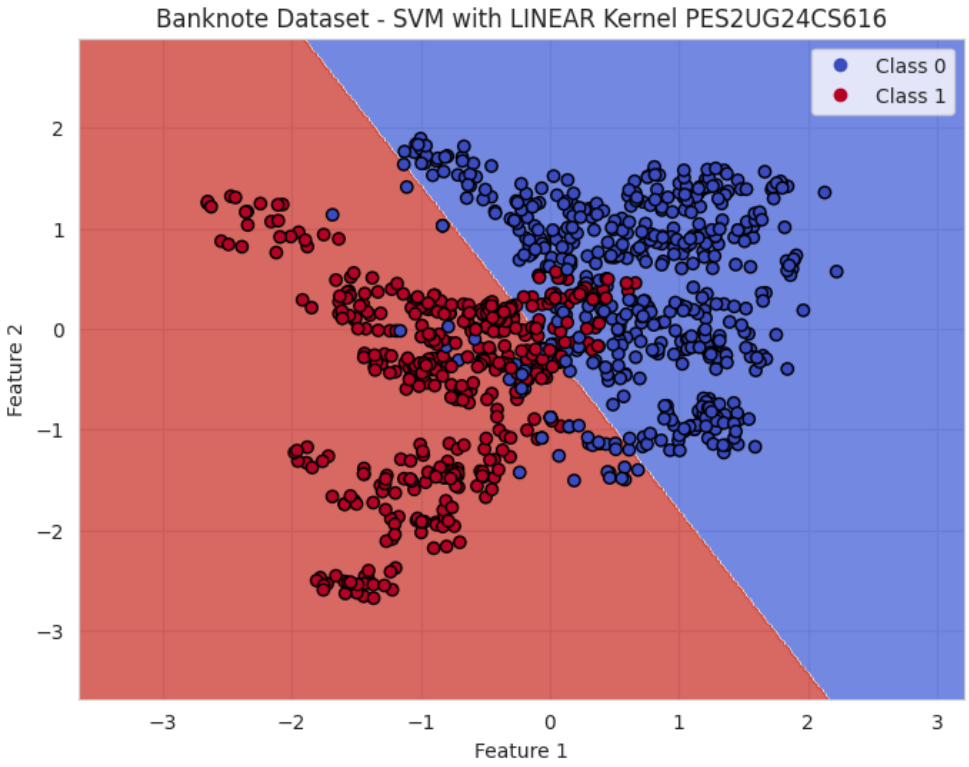
Screenshot 8: Moons Dataset - SVM with RBF Kernel



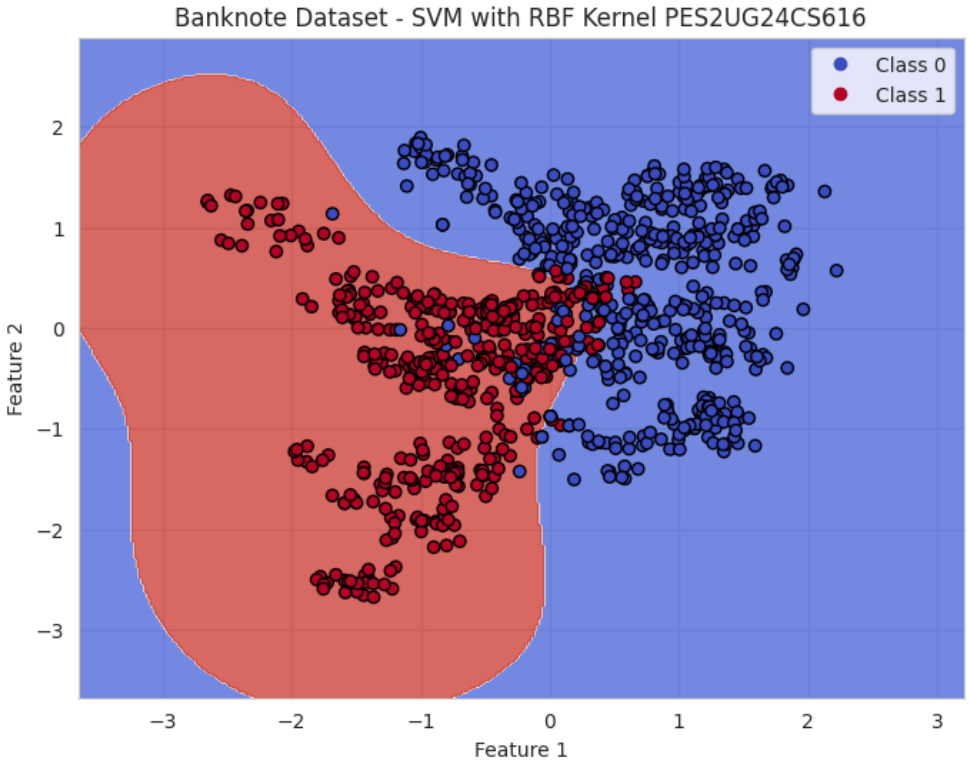
Screenshot 9: Moons Dataset - SVM with POLY Kernel



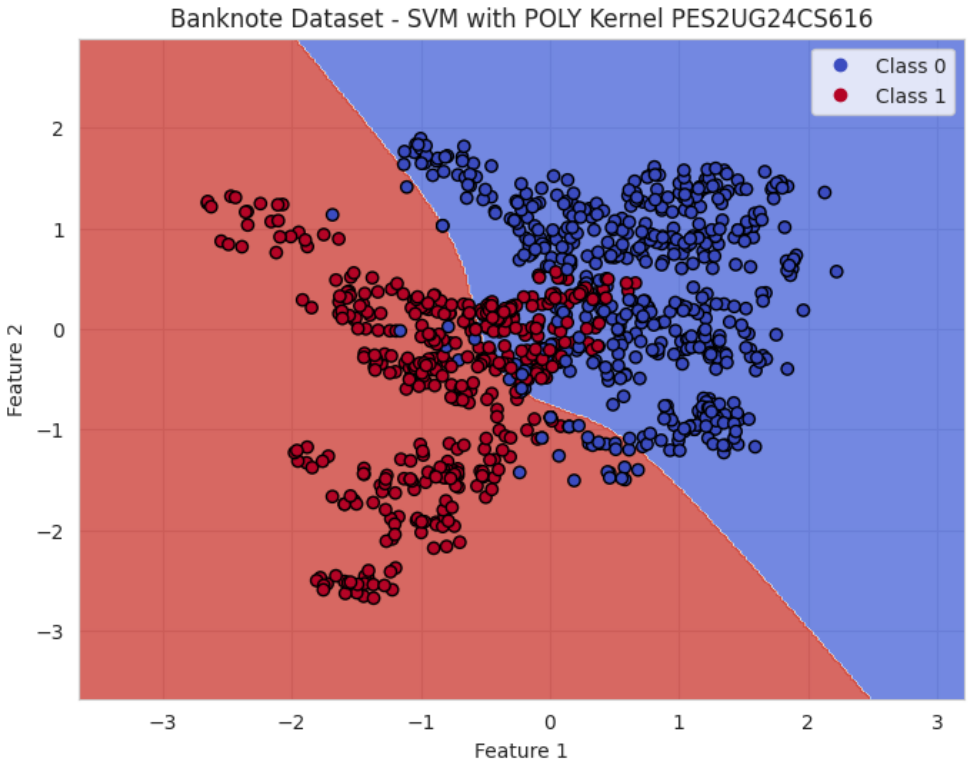
Screenshot 10: Banknote Dataset - SVM with LINEAR Kernel



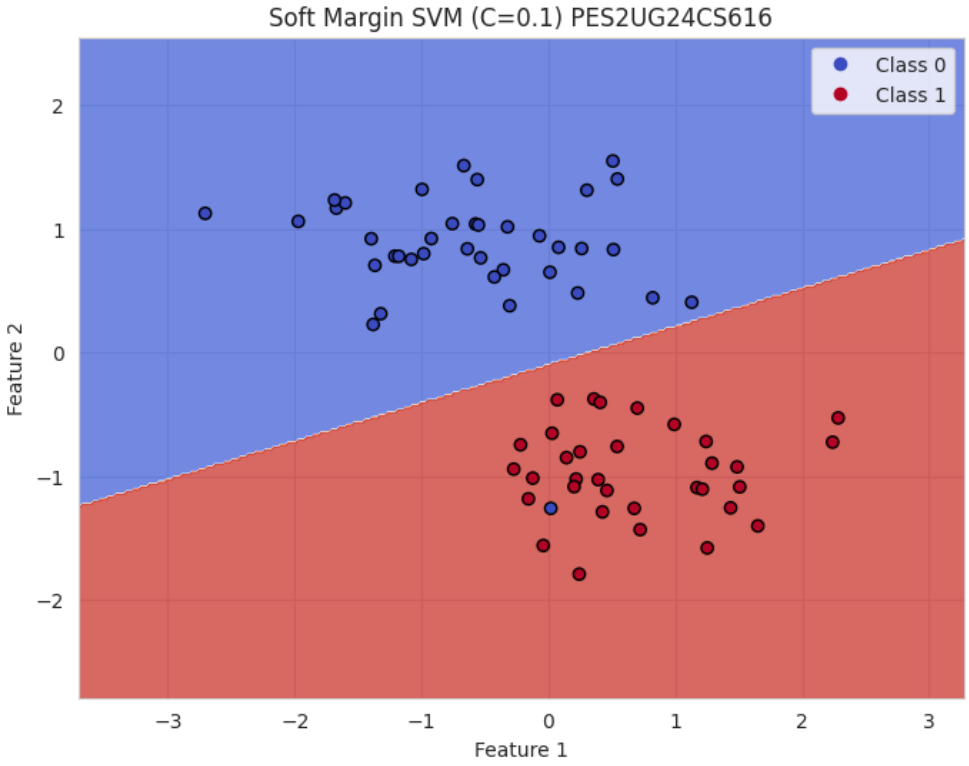
Screenshot 11: Banknote Dataset - SVM with RBF Kernel



Screenshot 12: Banknote Dataset - SVM with POLY Kernel



Screenshot 13: Soft Margin SVM (C=0.1)



Screenshot 14: Hard Margin SVM (C=100)

